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INSTITUTE OF TECHNOLOGY**



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PROJECT REPORT

(18CVP791)

on

**“ALTERNATIVE SUSTAINABLE MATERIAL DEVELOPED
UTILIZING GENERATED WASTE FOR DIVERSE APPLICATIONS”**

Submitted in partial fulfilment for the award of

Bachelor of Engineering (B.E)

in

“Civil Engineering”

Submitted by

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[UG Civil Engineering Program is Provisionally NBA (Tier I) Accredited up to 2023]

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CERTIFICATE

This is to certify⁸¹ a the Project Work entitled “**ALTERNATIVE SUSTAINABLE MATERIAL DEVELOPED UTILIZING GENERATED WASTE FOR DIVERSE APPLICATIONS**” carried out by **S RAMACHANDRAN, SHAIKH MUHAMMAD RISHAD, SRIRANJINI S B, TANZIM ULLA KHAN** bearing USN 1NT19CV055, 1NT19CV063, 1NT19CV071, 1NT19CV081 a bonafide student of Nitte Meenakshi Institute of Technology, Bengaluru in partial fulfilment of 7th Semester, B.E in Civil Engineering of the Visvesvaraya Technological University, Belgaum during the year 2022-2023. It is certified that all corrections/suggestions indicated have been incorporated in the report deposited in the department library. This Project Report has been approved as it satisfies the academic requirements as prescribed for the said degree.

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ABSTRACT

The use of single use plastic and the dependency on plastic for packing of various materials has led to over usage of plastic, the overall single use plastic generated worldwide is so large that the pollution that it causes onto the environment is humongous, hence we need to move towards replacing this single use plastic in our day to day lives even on a minute scale. This project offers to solve this problem to a small extent by creating a multipurpose product to replace the use of single use plastic. Through various testing and combination of materials this study aims to reduce the overall use of plastic all around the world while also utilizing the agricultural wastes into a useful product that is biodegradable, durable and reduces the environmental impact of the wastes.

ACKNOWLEDGEMENT

I would like to take this opportunity to thank a lot of eminent personalities, without whose constant encouragement, this endeavour of mine would not have become a reality.

Firstly, I express my deep gratitude to my guide **Mrs. Bharti Prasad and Dr. Megha Kulkarni**, Assistant Professors, Department of Civil Engineering, whose constant encouragement, guidance, and support throughout this work made me to carry out the I express my deep and sincere gratitude to Professor N. R Shetty, Director, Visionary of NMIT, Bengaluru, a kind-hearted person who is a Role Model for all the youngsters, for providing the infrastructure beyond curriculum encouraging research activities for the successful completion of Undergraduate studies. I take this opportunity to express my sincere words of gratitude to my beloved Dr. H C Nagaraj, Principal and Dr. V Sridhar, Dean Academic, for all the support and encouragement during my Undergraduate Studies. My sincere gratitude is due towards Management of my institution Nitte Meenakshi Institute of Technology, Bengaluru. Project work and I would like to express my sincere regards and heart full thanks to Dr. Bharathi Ganesh, Professor and Head of Civil Engineering Department.

I express my deep and sincere gratitude to Professor N. R Shetty, Director, Visionary of NMIT, Bengaluru, a kind-hearted person who is a Role Model for all the youngsters, for providing the infrastructure beyond curriculum encouraging research activities for the successful completion of Undergraduate studies. I take this opportunity to express my sincere words of gratitude to my beloved Dr. H C Nagaraj, Principal and Dr. V Sridhar, Dean Academic, for all the support and encouragement during my Undergraduate Studies. My sincere gratitude is due towards Management of my institution Nitte Meenakshi Institute of Technology, Bengaluru.

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DECLARATION

We, S RAMACHANDRAN, SHAIKH MUHAMMAD RISHAD, SRIRANJINI S ,TANZIM ULLA KHAN semester B.E in Civil Engineering bearing USN: 1NT19CV055, 1NT17CV063, 1NT19CV071, 1NT19CV081 of Nitte Meenakshi Institute of Technology Bengaluru, hereby declare that the work being presented in project entitled “**ALTERNATIVE SUSTAINABLE MATERIAL DEVELOPED UTILIZING GENERAL WASTE FOR DIVERSE APPLICATION**”, which is being submitted as partial fulfilment for VII semester B.E in Civil Engineering from Visvesvaraya Technological University Belagavi, is an authentic record of our project work that has been carried out by us during the academic year 2022-23 at **Nitte Meenakshi Institute of Technology**, Bengaluru under the guidance of **Mrs. Bharti Prasad and Dr.Megha Kulkarni**, Assistant professors, Department of Civil Engineering, Nitte Meenakshi Institute of Technology, Bengaluru

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CHAPTER 1

INTRODUCTION

Plastic has evolved into an indispensable part of all lives (Crespy, Bozonnet, & Meier, 2008). Plastic packaging is widely used in the protection and preservation of foods, cosmetics, medicines, and chemicals. But plastics have poor biodegradability. The poor biodegradability of plastics is due to the nature of the monomers and covalent bonds between them, which means they cannot be degraded by common physical, chemical, or enzymatic processes occurring in nature. As a result, plastic waste is typically disposed of by burying or burning, thereby causing pollution of land, water, and air that threaten human health and ecological environment. Plastic has grown at a compound annual rate of 8.4% since its mass production in 1950. Global plastic production reached around 348 million tons in 2017, which is almost 174 times higher than that in generated in 1950.

The production of plastics mainly comes from petrochemical engineering, which relies heavily on non-renewable petroleum resources. Therefore, the massive production of plastic not only causes serious pollution to the global environment, but also accelerates the depletion of petroleum resources. Plastic is reported to be a widespread pollutant in water sources, accounting for 60–95% of marine waste. (Schnurr et al., 2018). And it's estimated that around 8 trillion microbeads are released into wastewater every day (Schnurr et al., 2018). These plastics particles are getting into the bodies of marine animals and are passing along the food chain, even being found in larval fish (Gove et al., 2019). Plastics have also been reported to contaminate soil, whose health is critical to produce livestock and crops critical for human food security. Analysis of 29 flooded areas in Swiss nature reserves showed 90% of floodplain soils contained microplastic contaminants (Scheurer & Bigalke, 2018). Other studies have shown that microplastics in soil can be transferred to soil organisms (including worms or insects) and have harmful effects on their bodies (Chae & An, 2018). These microplastics may then be passed up the food chain when these smaller organisms are consumed by larger ones (such as birds and rodents).

Over the past decade growing demands for a healthier life by consumers has stimulated research on the relationship between the environment and human health (Fig. 1). The presence of plastics in fish has been shown to cause damage to their livers (Rochman, Hoh, Kurobe, & Teh, 2013) and immune systems (Limonta et al., 2019). In addition, a variety of additives in plastics have been shown to have serious effects on human health. They may

lead to human reproductive abnormalities and disruption of the endocrine and immune system, and even promote adult diabetes, early puberty, obesity and other chronic diseases (Kumar, 2018). One of the most effective strategies to reduce the problem of plastic pollution is to develop biodegradable materials that can replace it. At present, research on biodegradable materials mainly focuses on synthetic and natural polymers. Many synthetic polymers, including polylactic acid (PLA), polyvinyl alcohol (PVA), polycaprolactone (PCL), polybutyl succinic acid-butyl adipate (PBSA) and polyadipate butylene terephthalate (PBAT), are widely used biodegradable materials in daily life. In addition, several natural polymers are being used as raw materials for constructing biodegradable materials, such as proteins (whey protein, soy protein, silk fibroin), polysaccharides (chitosan, cellulose, starch), and lipids (beeswax, lauric acid).

Among the natural biodegradable materials, starch-based biodegradable materials are widely studied and applied. This is because starch is naturally abundant, low cost, non-toxic, renewable, biocompatibility and can form films. Moreover, the polyhydroxy structure of starch makes it easy to modulate its structure and functional properties using chemical or enzymatic methods.

Chemically, starch is a polymer consisting of long chains of glucose molecules held together by glycosidic linkages, and therefore has a chemical formula of $(C_6H_{10}O_5)_n$. At the molecular level, starch is composed of two main polymeric constituents (amylose and amylo-pectin), whose properties and ratios vary depending on the source (Chen, Zhang, et al., 2019). Natural starch generally exists in the form of granules (Fig. 2), which are spherical, oval, or irregular in shape, with diameters ranging from about 0.1 to 200 μm depending on the source and maturation stage (Pérez & Bertoft, 2010). Starch granules are insoluble in cold water, but if they are heated above a certain temperature more than water, they would absorb water and expand, and some of the amylose leaches from them (Liu et al., 2020). Moreover, the high temperature and excess water conditions of starch gelatinization do not relevant to all food uses of starch. At relatively low temperatures, the crystalline region of the starch granules maintains its integrity. But, if the starch suspension continues to be heated, the granules will continue to absorb water, and the crystalline regions will swell, eventually leading to the disruption of the starch granules and the formation of a hydrophilic colloidal solution (Lim, Wu, & Reid, 2000). This process is known as the gelatinization of starch. After gelatinization, when a starch solution is cooled to a sufficiently low temperature, starch molecules reorganize, hydrogen bonds reform, and ordered structures are re-established (Li, Hu, & Li, 2021). At the same time, the viscosity

of starch suspension increases and gel forms (Li et al., 2021). This process is known as retrogradation of starch. The structure and properties of starch are closely related to the formation and properties of starch-based biodegradable materials.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE REVIEW OF THE SELECTED JOURNAL PAPERS

1. Eco-Friendly (Watermelon Peels): Alternatives to Wood-based Particleboard Composites, 2011

Alternatives to Wood-based Particleboard Composites: The aim of this study was to investigate the suitability of using watermelon peels as alternatives to wood-based particleboard composites. 2011 Watermelon particles can be used as a substitute to wood-based particleboard for general purpose applications also besides being environmentally friendly of using watermelon and RLDPE in production of particleboard, this alternative to wood-based particleboard is very cost-effective. According to the end uses of wood-wastes and their possible reuse products, particleboard has found typical applications as flooring, wall and ceiling panels, office dividers, bulletin boards, furniture, cabinets, counter tops, and desktops and it seems that the manufacture of particleboard from biomass wastes is the most common way to reuse such waste materials Wood composite industries demand more wood raw material everyday even though the forest resources are diminishing.

The results obtained for the wear rate of the developed particleboard, from the results there is appreciable reduction in the wear rate of the particleboard with the addition of the RLDPE up to 60wt% for example: wear rate obtained are 0.56g at 30wt% and then to 0.31g at 60wt% RLDPE. The decrease in wear rate of the composites be attributed to higher load bearing capacity of hard particleboard and better interfacial bond between the particle and the matrix reducing the possibility of particle pull out which may result in higher wear.

Besides being environmentally friendly of using watermelon and RLDPE in production of particleboard, this alternative to wood-based particleboard is very cost-effective. The developed particleboard composites can be use in density particleboards for general purpose requirements like panelling, ceiling, partitioning etc.

2. Biodegradable packaging reinforced with plant-based food waste and by-products

Biodegradable packaging reinforced with plant-based food waste and by-products. Use of plastic food packaging has long been debated due to its negative environmental

impact. There is a potential to incorporate plant-based food waste or by-products as raw materials into biodegradable packaging for enhancement of packaging performance.

Being a large portion of the packaging materials, petroleum-based polymeric packaging is generally

nondegradable and only fewer than 10% of them are recycled. Reinforcing biodegradable packaging with PWBs and their effectiveness the common approach to fabricating packaging with PWBs in the most common way to prepare biodegradable films, materials from the PWBs were blended with biopolymers and additional additives, which can also come from PWBs, to form film solutions before casting to flat bottom holders for evaporation.

Improvement in antimicrobial and antioxidant capacities of packaging the antimicrobial properties of biodegradable packaging against typical gram-positive or gram-negative bacteria can also be improved by PWBs, especially in the form of extracts, such as pomegranate peel extract, black plum peel extract or pine needle extract. While some of the packaging properties still need to be improved to be competitive with plastics, in the next stage, innovations could also extend to other forms of packaging, such as disposable plates, wrapping paper, or multi-layer films.

3. Manufacturing fit-for-purpose paper packaging containers with controlled biodegradation rate by optimizing addition of natural fillers

Natural fillers were utilized for manufacturing horticultural packaging products. In case of paper sheets, 24 samples with dimension of 15 9 95 mm, were cut from each configuration of paper. Paper pots Paper pots were manufactured in the laboratory from the same pulp as reference flat paper sheets and with identical additives configuration. Polymerization degree Evaluation of paper properties the following tests were performed before degradation in soil to characterize the produced paper sheets: The degree of cellulose polymerization for paper products was determined on samples before and after soil degradation.

The high degree of scatter observed at 4 weeks exposure, together with minor cluster overlapping signifies a high heterogeneity within all tested papers as well as varied degradation kinetics, being dependent on the paper type. Effect of time on the biodegradation of the recycled paper with addition of 5% wheat bran in the forest soil the

vulnerability of the paper to degradation in soil was assessed by comparing the expected time to complete paper degradation. An Effect of the addition of pesticide on the biodegradation rate by micro fungi of paper articles supplemented with cereal bran and their breaking strength.

4.Effect of sugar palm nano fibrillated cellulose concentrations on morphological, mechanical, and physical properties of biodegradable films based on agro-waste sugar palm (Arenga pinnata (Wurmb Merr) starch

Sugar palm nanocrystalline cellulose reinforced sugar palm starch composite: degradation and water-barrier properties. Sugar palm fibres and sugar palm stem are by-products of the processed sugar palm fruits and sap for food products in Malaysia. Since sugar palm fibres are rich in cellulose, the findings from a few studies have showed that sugar palm fibres have huge potential as reinforcing component in various high-performance polymer composite applications increasing its commercial value as by-product from sugar palm cultivation. Sugar palm starch extraction and preparation the sugar palm starch was obtained from the stem of an amateur sugar palm tree. A suspension of sugar palm nanofibrillated cellulose, with an average length of about several micrometres and diameters of 5.5 0.99 nm were prepared from sugar palm fibres by high-pressure homogenisation method and was used to reinforce sugar palm starch matrix for the preparation of bio nanocomposites via solution-casting method

5. Husk as a substitute for Styrofoam plastic products manufacturing and packaging

The replacement of synthetic materials with rice husk material as manufacturing raw materials is used as a keyword to minimize the impact of environmental pollution.

While Styrofoam material is not decomposed by micro-organisms in the soil, so that the husk material can be used as a substitute for Styrofoam plastic materials. The physical-chemical properties of biology of Styrofoam synthetic materials are not easily degraded by micro-organisms, so that when the material is disposed of into waste, it will have implications for the damage to the soil, water, air and environmental health The global

community refuses to buy all products that still contain Styrofoam material, even though the presence of this synthetic material only functions as an electronic tool clamp in its packaging. Renewable Material Renewable materials are materials that can renew themselves naturally such as vegetation and animals.

6. Development of edible packaging film from banana peels and effect of glycerol concentration on the film properties

Edible film, banana peel, plasticizer, film properties Introduction Films are generally defined as stand-

alone thin layers of materials. Different properties of the developed films were studied, which includes

moisture content, tearing strength, film density, film solubility in water, film swelling index and water vapor transmission rate using following standard methods.

Determination of film density of the developed films was determined using densimeter of make and model Mettler-Toledo Sdn. Bhd. The immersing liquid used in this work was Xylene instead of distilled water to avoid water uptake by the hydrophilic film samples due to its lower density.

Determination of color difference Color difference was measured using the following formula $\Delta E = \text{square root of } 2$. Determination of film solubility for determining film solubility in water, three samples of 2 cm diameter were obtained from each film and dried in an oven at 105 °C for 24 h. The samples were weighed to determine the initial dry matter of each film. Water solubility of each film was determined by Determination of water sorption capacity Water sorption capacity of each film was studied by immersing a known mass of film sample into distilled water at room temperature.

Effect of glycerol concentration on film density Addition of plasticizer lowers the density of film as increasing concentration of plasticizer from 10 to 20%, slight decreasing in density of film from 2.812

to 0.743 g/cm³. Effect of glycerol concentration on the water absorption of edible film and its swelling index Films were emerged in water to verify the effect of plasticizer type and concentration on the hydrophilic nature of starch-based films.

7. Reprocessing Waste Cigarette Butts into Usable Materials

Objectives To recycle cigarette butts into usable materials to decontaminate the cigarette butt to reduce toxicity to produce fashionable products form cigarette butts to study the possibility of using the fiber material in butts for textile and clothing purpose.

Backpacks developed from waste cigarette butts by Terra Cycle Bricks Aeslina Abdul Kadir, and Abbas Mohajerani, had worked research on Possible Utilization of Cigarette Butts in Light Weight Fired Clay, and the results were very good that the work will encourage further investigation on the same topic.

Composition of cigarettes Different chemical treatments and processing's had been used to remove the bad sniff or odour of waste cigarette butts and to decrease the toxicity level of the innate chemicals found in the used cigarette butts.

8. Starch-based biodegradable packaging materials: A review of them preparation, characterization, and diverse applications in the food industry

Film solution casting. Film solution casting is a simple and well-established method to prepare film packaging materials, including paper.

The film solution casting method has been successfully used for the preparation of biodegradable films from corn starch, cassava starch and sweet potato starch. The application of film solution casting method in starch-based film industry is restricted.

Extrusion processing has been used for the preparation of hydroxypropyl starch-based films, corn starch-based films and cassava starch-based films. The addition of citric acid has been shown to reduce the water vapor transmittance of starch-based films and inhibit the degeneration Medina Jaramillo, Fama of films during storage.

Mylla ala starch-based film with glycerol had a looser structure than the film without glycerol, which resulted in lower water vapor and oxygen permeability, tensile strength and elastic modulus, and increased elongation at break.

9. Valorization and Application of Fruit and Vegetable Wastes and By-Products for Food Packaging Materials

Film solution casting Film solution casting is a simple and well-established method to prepare film packaging materials, including paper.

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10. Sustainable Use of Fruit and Vegetable By-Products to Enhance Food Packaging Performance

As a feasible alternative to reduce the production cost of edible films and to add value to food as a feasible alternative to reports reduce the the use production of edible films and to waste add value to food by-products, the literature also of fruit cost and vegetable processing to develop by-products, the literature also reports the use of fruit and vegetable processing waste to develop edible films or coatings.

The authors justified the better results of the soy protein-based film with the globular structure of the raw material and the likely lower protein-phenol interaction compared to the linear structure of fish gelatin film.

The total phenolic content of film stored at room temperature was around 26 to 50% higher than films at 4 and –18 C, respectively.

The results showed that in 31 days of testing, chitosan films enriched with olive by-products exerted a greater protective effect against peroxides formation compared to commercial and control chitosan films.

After selecting the best extraction method, the most interesting antioxidant extract was dispersed in an aqueous adhesive solution, which was used to develop two types of multilayer films, PE-based films, and PE/paper films.

The scavenger activity of the active films was evaluated by exposing the films to a gas rich in free radicals, and the best performance in terms of antioxidant activity was recorded for PE/PE. Recently, the extracts of olive leaves and pomace have also been applied to improve the properties of coating films.

Lignocellulose Nanofibers Obtained from Agricultural Wastes of Tomato, Pepper, and Eggplants Improves the Performance of Films of Polyvinyl Alcohol (PVA) for Food Packaging.

2.2 SUMMARY OF LITERATURE REVIEW

Table 2.1 Summary of literature reviews of all the research papers

SL.NO	NAME OF THE RESEARCH PAPER	NAME OF THE AUTHORS	NAME OF THE JOURNAL	YEAR OF PUBLICATION
1	Eco-Friendly (Watermelon Peels): Alternatives to Wood-based Particleboard Composites	U. D. IDRIS, V. S. AIGBODION, C. U. ATUANYA, ABDULLAHI. J	Tribology in industry	2011
2	Biodegradable packaging reinforced with plant-based food waste and by-products	Hongchao Zhang and Shyam Sablani	Current Opinion in Food Science	2021
3	Manufacturing fit-for-purpose paper packaging containers with controlled biodegradation rate by optimizing addition of natural fillers	Anna Sandak. Jakub Sandak. Izabela Modzelewska	Cellulose, Springer	2019

4	Effect of sugar palm nano fibrillated cellulose concentrations on morphological, mechanical, and physical properties of biodegradable films based on agro-waste sugar palm (<i>Arenga pinnata</i> (Wurmb Merr) starch	R.A. Ilyas a,b,*, S.M. Sapuana,b,*, Rushdan Ibrahimc , Hairul Abral d, M.R. Ishake , E.S. Zainudinb, M.S.N. Atikahf , N. Mohd Nurazzi a, A. Atiqahg, M.N.M.	Journal of Materials Research And Technology, Springer	2019
5	Husk as a substitute for styrofoam plastic products manufacturing and packaging	Reda Rizal , Lomo Mula Tua , Sargi Br. Ginting	Journal of Physics: Conference Series	2020
6	Development of edible packaging film from banana peels and effect of glycerol concentration on the film properties	Ram Prasad Bebartta, Komal Purushottam Katekhaye, Rameshwar Shivshankar Gajbharkar, Aishwarya Mishra and Nihar Ranjan Sahoo	International Journal of Chemical Studies	2019
7	Reprocessing Waste Cigarette Butts into Usable Materials	Jemal Assres,Bademaw Abate	International Journal on Textile Engineering and Processes	2018
9	Valorization and Application of Fruit and Vegetable Wastes and By-Products for Food Packaging Materials	Banu Bayram, Gulay Ozkan, Tina Kostka, Esra Capanoglu and Tuba Esatbeyoglu	Molecules, MDPI	2021

10	Sustainable Use of Fruit and Vegetable By-Products to Enhance Food Packaging Performance	Flavia Dilucia, Valentina Lacivita, Amalia Conte * and Matteo A. Del Nobile	Foods,MDPI	2020
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CHAPTER 3

AIM AND SCOPE OF THIS STUDY

1. There is a shift towards biodegradable packaging mainly attributed to the rise in health awareness and environmental concerns associated with biodegradable packing.
2. In India packaging industry is expected to grow from INR - 2372.56 billion in 2015 to INR -5412.40 billion in 2022 at a CAGR of 18% (source -economic times). This shows the growing demand for packaging material. Therefore, the market potential for biodegradable packaging materials is also highlighted.
3. Globally the food packaging market is anticipated to grow at a CAGR of 4.4% till 2019 (Source -global for packaging market 2015-19, Technavio). According to the report by Wise guy reports the bio- degradable for packaging market is forecasted to increase from INR -348.47 billion in 2015 to INR -1060.24 billion in 2022 at CAGR of 17.2%.
4. The European Bioplastics association reported that the global production capacity of biodegradable polymers reached 428000 metric tons in 2010. The bio plastic industry is noted to have solid growth in PLA, PHA and plant starch-based material. Bio based polymer production would triple from 5.1million tons in 2013 to 17 million tons in 2020 (Darby,2012).
5. From a waste management perspective, high value, multilayer, biodegradable food packing might be a useful replacement for current multilayered packing that is non-recyclable and Nondegradable.

CHAPTER 4

OBJECTIVES

1. To collect biodegradable and non-biodegradable waste from all sections of community to recycle them.
2. Characterization of constituent wastes to produce alternative materials for diverse applications.
3. To produce a quality product from generated waste to reduce the environmental impact.
4. From a waste management perspective, high value, multilayer, biodegradable food packing might be a useful replacement for current multilayered packing that is non-recyclable and Nondegradable
5. To propose a material which is easy to fabricate, safe and cost-effective solution.

CHAPTER 5

METHODOLOGY

The following methods are used for preparation of the alternative materials,

5.1 FILM AND FOAM PROCESSING USING STARCH

The procedure casting film and foam using starch is as follows

5.1.1. INVESTIGATION: Over the past decade growing demands for a healthier life by consumers has stimulated research on the relationship between the environment and human health variety of additives in plastics have been shown to have serious effects on human health. They may lead to human reproductive abnormalities and disruption of the endocrine and immune system, and even promote adult diabetes, early puberty, obesity, and other chronic diseases. One of the most effective strategies to reduce the problem of plastic pollution is to develop biodegradable materials that can replace it. Among the natural biodegradable materials, starch-based biodegradable materials are widely studied and applied. This is because starch is naturally abundant, low cost, non-toxic, renewable, biocompatibility and can form films. Moreover, the polyhydroxy structure of starch makes it easy to modulate its structure and functional properties using chemical or enzymatic methods.

5.1.2. MATERIALS USED: starch obtained from various natural sources like corn, cassava, sweet potato, and many more natural occurring substances

5.1.3. FILM SOLUTION CASTING: Film solution casting is a simple and well-established method to prepare film packaging materials, including paper.

Initially, a film-forming solution is prepared by heating and stirring a mixture of film-forming materials and solvent, which is a film suspension of gelatinized starch. This solution is then poured onto a smooth surface, such as a glass plate or plastic plate. The spread solution is then dried in an oven or at room temperature to evaporate the solvent. Finally, the prepared films are removed from the plate by peeling. It is advisable to carry out degassing of the film suspensions of gelatinized starch before pouring on to the plate to avoid the presence of air bubbles in the final film. The film solution casting method has been successfully used for the preparation of biodegradable films from corn starch cassava starch and sweet potato starch.

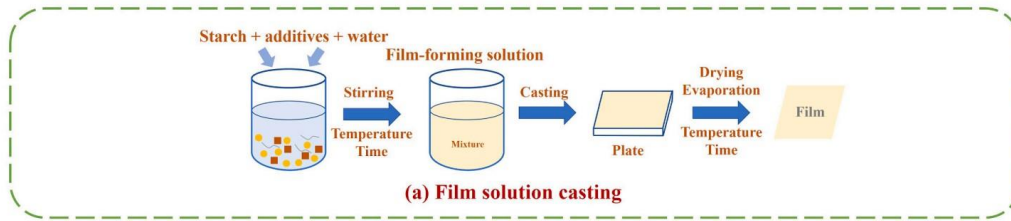


Fig 5.1 Film casting process

5.1.4. FOAMING PROCESS: Foaming processing is a method of producing starch-based foams. Four main approaches have been developed to create these foams: baking, moulding, extrusion, and supercritical fluid foaming. Baking foaming involves adding a mixture of starch, foaming agent, and other additives into a mold and then heating in an oven. During baking, evaporation of water vapor leads to foam formation, and the viscosity of the starch paste plays an important role in the nature of the gas bubbles formed. Moulding foaming is a method like baking foaming. In this case, the foaming material is placed in a temperature-controlled hydraulic press and a foam is formed by heating and pressure changes. Biodegradable cassava starch-based foams mixed with grape stalks have been formed using this approach, and the prepared foam was shown to

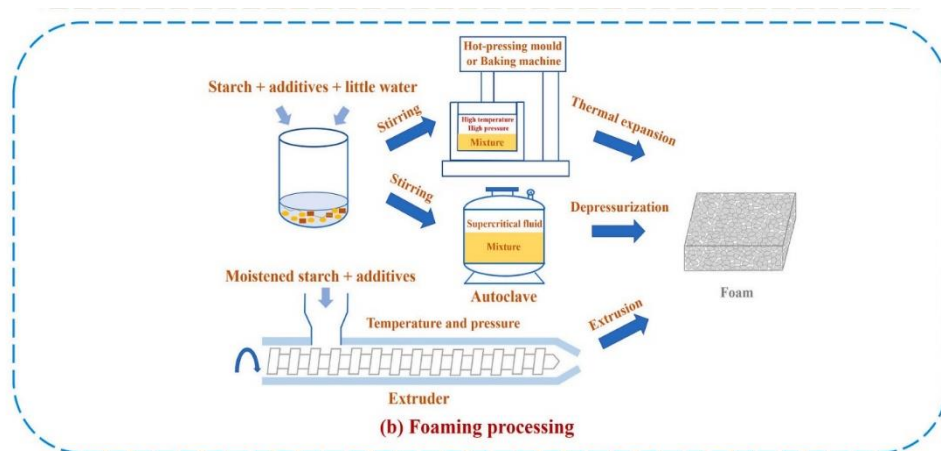


Fig 5.2 Foaming process

have potential for use as a food packaging material. Supercritical fluid foaming typically uses supercritical carbon dioxide to form gas bubbles. The foaming material and carbon dioxide are mixed under pressure. The device is then depressurized, causing the carbon dioxide gas to form bubbles leading to the creation of a foamed product.

5.1.5. EXTRUSION PROCESS: Extrusion blowing involves blowing, pressing, and cooling the extruded film-forming material to produce films. Extrusion compression molding forms the films through hot pressing and cooling the extruded film-forming material. Extrusion injection molding is like extrusion compression molding since both methods include extruding and molding during processing. But extrusion injection molding is the molten raw materials injected into the mold, such as cooling to obtain the product. Extrusion injection molding is widely used in polymer processing because it can produce many complex plastic products at low-cost Extrusion

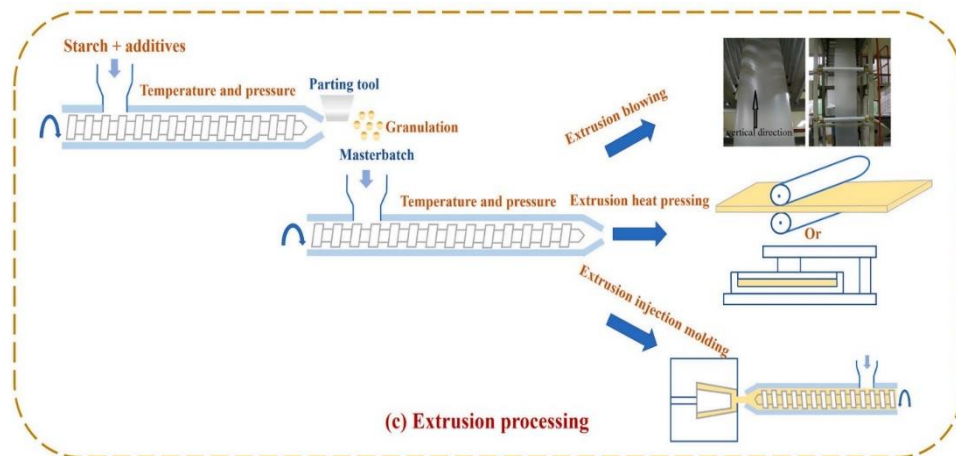


Fig 5.3 Extrusion Processing

has been used for the preparation of hydroxypropyl starch-based films corn starch-based films and cassava starch-based films. The film extrusion process has a high production efficiency and the films produced have morphologies like those produced using the casting method.

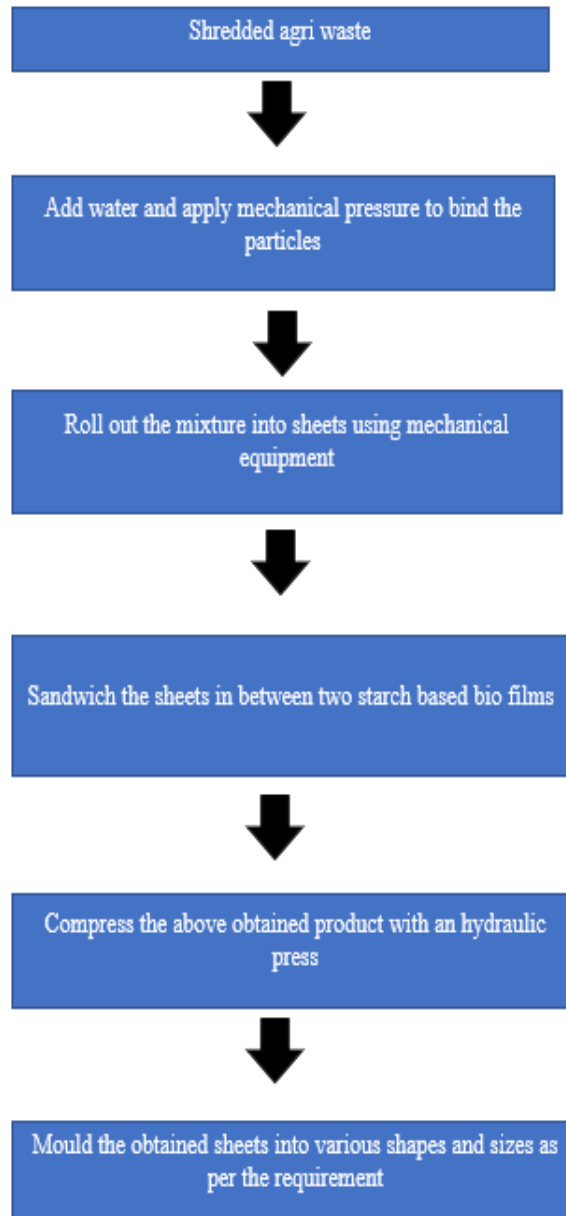
5.1.6. CONCLUSION: Starch-based biodegradable materials may have an important role to play in the development of more environmentally friendly and sustainable food packaging materials in the future. One of the current limitations is that the retrogradation of starch results in poor mechanical properties. To solve this problem, various approaches have been investigated, such as chemical, enzymatic, or physical modification of starch, as well as the addition of other biopolymers and functional additives to the films. Although some of these approaches have led to improvements in film properties, a great deal of work is still required to create starch-based films with similar optical, barrier and mechanical properties as those formed from synthetic polymers.

5.2 RICE HUSK DRAWN INTO SHEETS USING MECHANICAL PROCESS:

Rice husk is grounded into micro-fibrous / nano fibrous particles and mixed with water to make a heterogenous mixture. This mixture is fed into the mechanical pressure chamber to bind the materials together. The bonded mixture is drawn into thin sheets using a roller and dried. Further the sheet which is rolled out is sandwiched between two starch-based bio films by compressing it using a hydraulic press. The obtained sheets are casted into different molds to produce the packaging product of required size and shape.

This three-layered package caters for durable and sustainable packaging as rice husk increases the strength of the package, resists fungal decomposition and moisture penetration, insulates well and is biodegradable. Rice husk contains cellulose (50%), lignin (25%–30%), silica (15%–20%) which counteracts for the poor mechanical properties, low water vapor barrier property, and low freeze stability of the starch while the sheet drawn out from rice husk is sandwiched between two layers of the starch-based bio films.

METHODOLOGY FLOWCHART



CHAPTER 6

MATERIALS

The materials used for the preparation of film are as follow

6.1 Starch

Starch obtained from various natural sources like corn, cassava, potato, and many more natural occurring substances



Fig 6.1 Starch obtained from potato

Potato starch and its derivatives have special properties, such as a low gelatinization temperature and a high paste consistency. Potato starch is preferred in the food industry,²⁹ because its pastes have a good clarity (due to a small amount of lipids and protein) and a neutral flavour.

6.2 Rice husk ash and Sugarcane Bagasse



Fig 6.2 Rice husk

The average chemical composition of rice husk as reported is 20% silica, 22% lignin, 38% cellulose, 18% pentosans, and small amount of other organic matters. The high percentage of silica, which is very unusual within nature, and this intimate blend of silica and lignin make the rice husk resistant to water penetration and fungal decomposition.

6.3 Banana peel

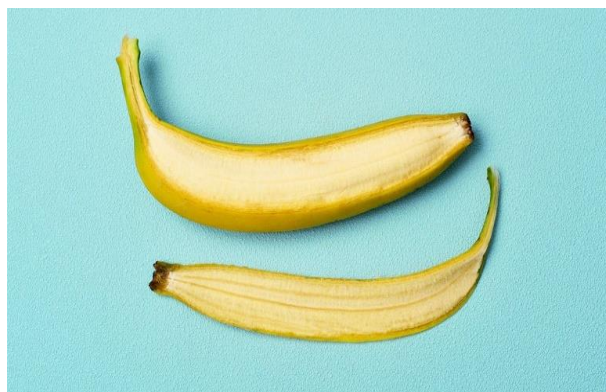


Fig 6.3 Banana



Fig 6.4 Uses of Banana

Banana peels provide structural stability and prevents mechanical damage during transportation, handling, and display.

It has good adhesion to surface of food to be protected providing uniform coverage. It controls water migration both in and out of protected food to maintain desired moisture content. It provides semi-permeability to maintain internal equilibrium of gases involved in aerobic and anaerobic respiration, thus retarding senescence. It prevents loss or uptake of components that stabilize aroma, flavour, nutritional and organoleptic characteristics necessary for consumer acceptance while not adversely altering the taste or appearance.

6.4 Glycerol



Fig 6.5 Glycerol

Glycerol is a colourless, hygroscopic, highly viscous syrupy liquid. It is miscible with water and alcohol in all proportions. The boiling point of glycerol is 563 K which is quite high. Glycerol is non-toxic in nature. Glycerol is hygroscopic, allowing it to absorb and retain water. This property is particularly useful in food and cosmetic preparations.

6.5 Vinegar



Fig 6.6 Vinegar

Both grain vinegars and fruit vinegars, which are fermented by traditional methods, possess a variety of physiological functions, such as antibacterial, anti-infection, antioxidation, blood glucose control, lipid metabolism regulation, weight loss, and anticancer activities.

6.6. Pomegranate Peels



Fig 6.7 Pomegranate Peels

Recent studies have demonstrated the potent antioxidant and antimicrobial effects of using pomegranate rind and seed as natural food additives, thus making researchers incorporate them into bioplastics and edible coatings for food packaging. Additionally, these components have shown great plasticizing effects on packaging materials while extending the shelf life of food through active packaging.

6.7 BETEL LEAVES

Leaves contained caryophyllene, cadinene, γ -lactone, allyl catechol, p-cymene and eugenol methyl ether in varying amounts beetle leaves and the purified compounds have been found to possess important antiseptic (e.g., oral hygiene), anti- diabetic, cardiovascular, anti-inflammatory/ immunomodulatory, antiulcer, hepatic protective, and anti-infective properties [and are also active against many species of bacteria and fungi



Fig 6.8 Betel leaves

CHAPTER 7

FILM CASTING PROCESS

- Take 100g of starch powder obtained from potato
- Add to water and mix it thoroughly in a utensil
- Add 5ml of glycerol and 5 ml of vinegar to the mixture
- Heat the mixture at nearly 90 to 100 degrees Celsius for around 5-10 minutes until it forms a sticky jelly substance
- After heating the mixture pour the mixture onto the aluminium foil/glass plate and roll it with the help of roller to spread the mixture onto the foil equally
- Then allow the mixture to settle and dry it using oven drying for about 10 to 15 minutes
- After drying it, try to remove the created film layer slowly from the foil/plate



Fig 7.1. Mixture of starch and water



Fig 7.2 Mixture of Starch and Rice Husk in water used for casting

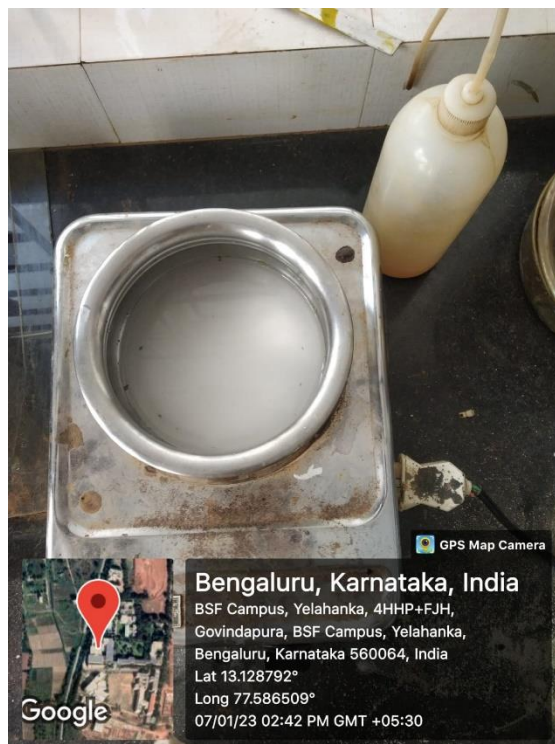


Fig 7.3 Mixture of Starch and betel leaves in water used for casting



Fig 7.4. Mixture of rice husk and Starch powder heated along with Vinegar and Glycerol



Fig 7.5 A mixture of Starch, betel leaf, glycerol, and vinegar heated using a stove for film casting



Fig 7.6 The starch mixture spread evenly onto the plate to form thin film layer



Fig 7.7 Starch and rice husk film casted onto a plate and kept for drying



Fig 7.8. The Starch mixture spread on the foil evenly with the help of a mechanical roller

CHAPTER 8

PRODUCT TESTING

- The following tests are required to be performed to evaluate the physical properties of the film:
- Degradation in water – the film degrades in water in a time span at which each constituent takes its time to degrade
- Degradation in soil- decomposition of the film takes minimum of 20 days in the natural conditions
- Odour – test for objectionable odour imparted by the film
- Microbial and fungal resistance
- Heat resistance test
- Durability test
- Elasticity test

CHAPTER 9

RESULTS AND CONCLUSION

9.1 RESULTS

The results obtained for the film casting process using starch, glycerol and vinegar are as follows:

- The film casted onto the plate has uniform thickness and it is thin
- The film casted onto the aluminum foil was spread using a roller and has uneven thickness
- Both the films casted could be drawn into thin sheets after oven drying
- The results obtained for the film casting process using starch and rice husk,
- Shows better elastic properties compared to the film casted in the previous trial
- Exhibits durability and strength
- Shows resistance against wear and tear

The various following tests conducted on material to know their suitability as packaging materials for diverse applications:

The below tests are physically verified in the laboratory and for the accurate results the samples must be given for the instrument tests to the CMTI Bangalore

Table 9.1 List of experiments to obtain the physical properties of the film

SL. No	Name of the test	Results
1	Fire resistance	It has resistance against fire
2	Degradation in water	Degrades in water with the span of 20 days for the film fabricated using potato starch.
3	Degradation in soil	Shows partial decomposition when kept in soil
4	Odour	No objectionable odour that hinders the film

5	Microbial and fungal resistance	There is no evidence of microbial growth after a span of 20 days
6	Heat resistance	Resistance against boiling temperature 100 degrees
7	Durability	Retains the same shape and strength after 20 days of casting
8	Elasticity	Examined physically and shows good resistance to elasticity

9.2 CONCLUSION

- The biodegradable waste from marketplaces is successfully collected and constituents are extracted
- The various constituents are selected based on their characteristics
- The various trials have been made to produce good quality product
- The produced product is showing satisfactory results to be used as a packaging material
- Since the fabricated material can be easily decomposed either in soil or water, henceforth it will be good alternative for existing single use plastic which is extensively used

CHAPTER 10

SCOPE FOR FUTURE STUDIES

To improve the strength of the above obtained film we can use the following ingredients:

- Rice husk –A fine mechanically powdered Rice husk can be added to the mixture of starch and water by which the starch binds the rice husk to improve overall strength as the fibrous property of the Rice husk helps in increasing the durability of the packaging material
- Banana peel-Add properly cut banana peels onto the starch mixture then follow the same procedure that has been used for film casting.
- Peels of Beetroot, Potato and Brinjal can be used along with the starch obtained from Potato to form the mixture
- Pomegranate peels and extract can be added to the mixture to improve anti-microbial activity.
- Powdered banana leaves, patravali leaves and betel leaves added to the mixture helps in increasing the anti-microbial and anti-fungal activities the film

Table 10.1 List of ingredients that can be used for future experiments

SL.NO.	INGREDIENTS	PURPOSE
1	Rice husk	To improve the strength and durability of the film
2	Banana peel	To improve the overall quality of the film, possesses good microbial property and is a good antioxidant
3	Beetroot peel, Brinjal peels	Anti-inflammatory and good antioxidant properties
4	Pomegranate peels	Rich in fibers antimicrobial property
5	Betel leaves, Patravalli leaf, Banana leaf	Produces good bonding strength and is anti-microbial

CHAPTER 11

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